A REGULAR LOCAL ALGEBRA WITH SEPARABLE RESIDUE FIELD EXTENSION IS FORMALLY SMOOTH OVER THE BASE FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Formal smoothness can be checked after completion. Lemma \ref{lemma-formally-smooth-completion}, cited at proof lines 9633-9634 and 9650-9651. Reduces to a complete local algebra and later passes from the polynomial ring to the power series ring.
- A complete regular local algebra over a field is a power series ring over its residue field. Lemma \ref{lemma-power-series-over-residue-field}, cited at proof lines 9636-9637. Identifies the algebra in the complete case.
- Formal smoothness is stable under composition, and a polynomial algebra is formally smooth over its base. Algebra, Lemma \ref{algebra-lemma-compose-formally-smooth}, cited at proof lines 9643-9644. Gives formal smoothness of the polynomial ring over the base field.
- Formal smoothness upgrades from the algebraic sense to formal smoothness for the adic topology. Lemma \ref{lemma-formally-smooth}, cited at proof lines 9645-9647. Puts the polynomial and power series rings into the adic topology used in the statement.

Source reference. Original lemma 07EJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9449–9449. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. Navigation pointer to a statement the

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Source reference. Original lemma 0C35: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15180–15180. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 031H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3984–3984. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9442–9442. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let k be a field. Let $(A, \mathfrak{m}, K)$ be a regular local k-algebra such that K/k is separable. Then $k \to A$ is formally smooth in the $\mathfrak{m}$-adic topology.*

Proof. It suffices to prove that the completion of A is formally smooth over k, see Lemma 07ED. Hence we may assume that A is a complete local regular k-algebra with residue field K separable over k. By Lemma 0C35 we see that $A = K[[x_1, \ldots, x_n]]$.

The power series ring $K[[x_1, \ldots, x_n]]$ is formally smooth over k. Namely, K is formally smooth over k and $K[x_1, \ldots, x_n]$ is formally smooth over K as a polynomial algebra. Hence $K[x_1, \ldots, x_n]$ is formally smooth over k by Algebra, Lemma 031H. It follows that $k \to K[x_1, \ldots, x_n]$ is formally smooth for the $(x_1, \ldots, x_n)$-adic topology by Lemma 07EC. Finally, it follows that $k \to K[[x_1, \ldots, x_n]]$ is formally smooth for the $(x_1, \ldots, x_n)$-adic topology by Lemma 07ED. □